

1. Configuration du Serveur Samba (Ubuntu)

Pour centraliser les données de sauvegarde, nous avons configuré un partage réseau via le protocole Samba sur un serveur Ubuntu.

- **Commandes d'installation et de préparation :**

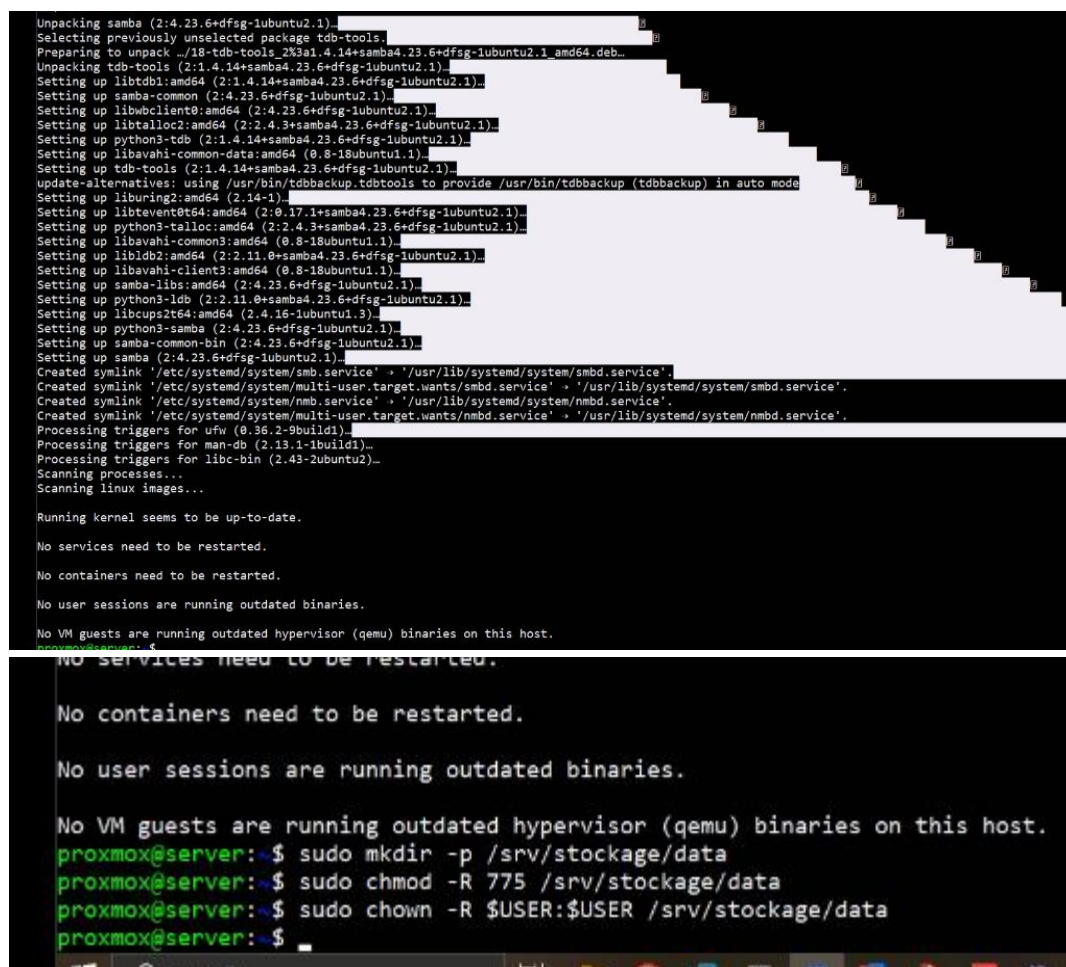
```
sudo apt update && sudo apt install samba -y
```

```
sudo mkdir -p /srv/stockage/data
```

```
sudo chown -R proxmox:proxmox /srv/stockage/data
```

```
sudo chmod -R 775 /srv/stockage/data
```

```
sudo smbpasswd -a proxmox
```



```
Unpacking samba (2:4.23.6+dfsg-1ubuntu2.1)...
Selecting previously unselected package tdb-tools.
Preparing to unpack ./18-tdb-tools_2K3a1.4.14+samba4.23.6+dfsg-1ubuntu2.1_amd64.deb...
Unpacking tdb-tools (2:1.4.14+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up libtdb1:amd64 (2:1.4.14+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up samba-common (2:4.23.6+dfsg-1ubuntu2.1)...
Setting up libwbclient0:amd64 (2:4.23.6+dfsg-1ubuntu2.1)...
Setting up libtalloc2:amd64 (2:2.4.3+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up python3-tdb (2:1.4.14+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up libavahi-common-data:amd64 (0.8-18ubuntu1.1)...
Setting up tdb-tools (2:1.4.14+samba4.23.6+dfsg-1ubuntu2.1)...
update-alternatives: using /usr/bin/tdbbackup.tdbtools to provide /usr/bin/tdbbackup (tdbbackup) in auto mode
Setting up liburing2:amd64 (2.14-1)...
Setting up libevent0:amd64 (2:8.17.1+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up python3-talloc:amd64 (2:2.4.3+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up libavahi-common3:amd64 (0.8-18ubuntu1.1)...
Setting up libldb2:amd64 (2:2.11.0+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up libavahi-client3:amd64 (0.8-18ubuntu1.1)...
Setting up samba-lsbs:amd64 (2:4.23.6+dfsg-1ubuntu2.1)...
Setting up python3-ldb (2:2.11.0+samba4.23.6+dfsg-1ubuntu2.1)...
Setting up libcups2t64:amd64 (2.4.16-1ubuntu1.3)...
Setting up python3-samba (2:4.23.6+dfsg-1ubuntu2.1)...
Setting up samba-common-bin (2:4.23.6+dfsg-1ubuntu2.1)...
Setting up samba (2:4.23.6+dfsg-1ubuntu2.1)...
Created symlink /etc/systemd/system/smb.service → /usr/lib/systemd/system/smbd.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/smbd.service → /usr/lib/systemd/system/smbd.service'.
Created symlink /etc/systemd/system/nmb.service → /usr/lib/systemd/system/nmbd.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/nmbd.service → /usr/lib/systemd/system/nmbd.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
proxmox@server:~$ sudo mkdir -p /srv/stockage/data
proxmox@server:~$ sudo chmod -R 775 /srv/stockage/data
proxmox@server:~$ sudo chown -R $USER:$USER /srv/stockage/data
proxmox@server:~$
```

- **Configuration du partage :** Le fichier /etc/samba/smb.conf a été édité pour inclure la définition du partage [Stockage-PFE] :

[Stockage-PFE]

```
path = /srv/stockage/data
```

```
read only = no
```

```
browsable = yes
```

```
writable = yes
```

```
# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
    comment = Printer Drivers
    path = /var/lib/samba/printers
    browseable = yes
    read only = yes
    guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[Stockage-PFE]
    path = /srv/stockage/data
    browsable = yes
    writable = yes
    guest ok = no
    read only = no_
```

- Redémarrage des services :

sudo systemctl restart smbd nmbd

```
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
proxmox@server:~$ sudo mkdir -p /srv/stockage/data
proxmox@server:~$ sudo chmod -R 775 /srv/stockage/data
proxmox@server:~$ sudo chown -R $USER:$USER /srv/stockage/data
proxmox@server:~$ sudo nano /etc/samba/smb.conf
proxmox@server:~$ sudo smbpasswd -a anas
New SMB password:
Retype new SMB password:
Failed to add entry for user anas.
proxmox@server:~$ sudo systemctl restart smbd nmbd
proxmox@server:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-06-25 03:12:56 UTC; 23s ago
   Invocation: ed4602f23b71483b8aa8273ffb58a3f5
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
   Process: 13941 ExecCondition=/usr/share/samba/is-configured smb (code=exited, status=0/SUCCESS)
   Process: 13944 ExecStartPre=/usr/share/samba/update-apparmor-samba-profile (code=exited, status=0/SUCCESS)
  Main PID: 13954 (smbd)
    Status: "smbd: ready to serve connections..."
     Tasks: 3 (limit: 1827)
    Memory: 7.7M (peak: 7.7M)
       CPU: 114ms
   CGroup: /system.slice/smbd.service
           └─13954 /usr/sbin/smbd --foreground --no-process-group
             └─13957 "smbd: notifyd" .
               └─13958 "smbd: cleanupd "
```

2. Intégration dans l'hyperviseur Proxmox

Une fois le partage opérationnel, nous l'avons lié à l'interface Proxmox VE pour gérer les sauvegardes.

- **Ajout du stockage** : Via l'interface *Datacenter* > *Storage* > *Add* > *SMB/CIFS*, le volume a été configuré avec les accès utilisateur proxmox.
- **Vérification de la connectivité** : Les commandes suivantes ont permis de valider le montage et les droits en écriture sur le volume :

PROXMOX Virtual Environment 9.2.3

Server View: Datacenter (virtacore-dc1)

Node: pve1

Package versions

Summary

pve1 (Uptime: 02:30:49)

CPU usage: 4.42% of 4 CPUs

Load average: 0.27, 0.17, 0.11

IO delay: 0.00%

RAM usage: 87.97% (3.32 GiB of 3.78 GiB)

KSM sharing: 235.79 MB

/ HD space: 41.60% (18.61 GiB of 44.73 GiB)

SWAP usage: 2.98% (121.99 MB of 4.00 GiB)

CPU(s): 4 x AMD Ryzen 5 PRO 4650U with Radeon Graphics (2 Sockets)

Kernel Version: Linux 7.0.12-1-pve (2026-06-09T21:07Z)

Boot Mode: Legacy BIOS

Manager Version: pve-manager/9.2.3/d8bde103346c88a

Repository Status: Proxmox VE updates Non production-ready repository enabled

CPU Usage

Cluster log

Start Time	End Time	Node	User name	Description	Status
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PROXMOX Virtual Environment 9.2.3

Server View: Datacenter (virtacore-dc1)

Health

Status: Cluster: virtacore-dc1, Quorate: Yes

Nodes: Online 2, Offline 0

Guests

Virtual Machines: Running 1, Stopped 1, Templates 6

LXC Container: Running 0, Stopped 0

Cluster log

Start Time	End Time	Node	User name	Description	Status
Jun 25 04:48:30	Jun 25 04:48:57	pve1	root@pam	Shell	OK

ID	Type	Content	Path/Target	Shared	Enabled	Bandwidth Limit
local	Directory	Backup, Import, ISO image, Snippets, Cont...	/var/lib/vz	No	Yes	
local-lvm	LVM-Thin	Disk image, Container		No	Yes	
volume	SMB/CIFS	Disk image, Import, ISO image, Container, ...	/mnt/pve/volume	Yes	Yes	

Add: SMB/CIFS

General Backup Retention

ID: Stockage-NAS

Server: 192.168.100.90

Username: proxmox

Password: *****

Share: Stockage-PFE

Subdirectory: /some/path

Nodes: All (No restrictions)

Node: pve1

Memory usage: 87.1 %

CPU usage: 2.2% of 4 ...

Node: pve2

Memory usage: 42.9 %

CPU usage: 4.7% of 4 ...

Preallocation: Default

Allow Snapshots as Volume-Chain

Snapshots as Volume-Chain are a technology preview.

Help Advanced Add

local-lvm LVM-Thin Disk image, Container

Add: SMB/CIFS

General Backup Retention

ID: Nodes:

Server: Enable: ☐

Username: Content:

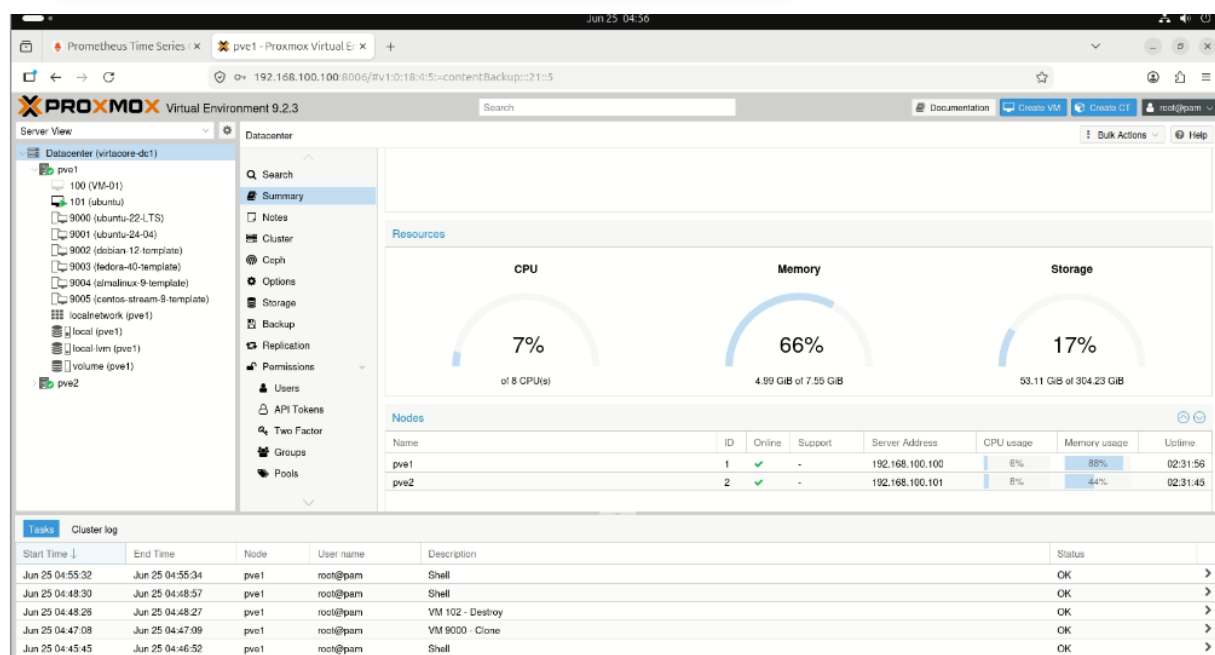
Password: Domain:

Share: Subdirectory:

Preallocation: ☐ Allow Snapshots as Volume-Chain

Snapshots as Volume-Chain are a technology preview.

☒ Advanced



3. Conclusion

Cette configuration assure une gestion centralisée des ressources (Backups, ISO, Templates) et garantit la haute disponibilité des données, répondant ainsi aux exigences de notre architecture projet.